Index sheet for Course file

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4	Timetable (of all sections of that course)	-
5	Student list (template – 3) (extracted from SAP)	-
6	Continuous Internal Evaluation (CIE)	-
	- Question paper for the CIE components (min – n +1; max – n + 4) To be common across all sections	-
	- Scheme of evaluation (common) and rubrics of all components	-
	- 2 sample answer scripts/assignments of the students (of each CIE component) (one best script, one average script)	-
	- Consolidated marks sheet of CIE (of all components)	-
	- List of slow learners (students whose score is less than 40% in a CIE will be termed as slow learners)	-
	Remedial class details - Time table of the class to be conducted - Classes work done statement - Retest (Question paper and answer sheets) (which includes the details of the classes taken, retest conducted and the improved marks from the retest to be considered for CIE)	-
	- Finalized CIE marks signed by all the students (which is submitted to the exam office)	-
7	- SA & FI list	-
	- Final attendance sheet (mentioning the % of class attended)	-
8	Semester End Examination (SEE)	-
	- Question paper	-
	- Scheme of evaluation	-
	- Evaluation rubrics	-
	- Result sheet of CIE	-

9	Course-end survey (questions to be specific to the course; not more than 8)	-
10	Course attainment	-

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Course outline (Syllabus of the course) (Template – 1)

Course Code: CS2228 **Name of the Programme: B.Tech (Hons.)** **Name of the Course: Deep Learning** **Semester: V**		
Course credit	**No. of hours per week**	**Total no. of Teaching hours**
03	**2+0+2**	**60**
Course Objectives:		
● To introduce the foundational concepts of neural networks and their application in classification tasks.		
● To explore advanced deep learning architectures such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for solving computer vision and sequence modeling problems.		
● To develop practical skills in optimizing and deploying deep learning models using industry-standard SDKs such as OpenVINO, SNPE, and TensorRT.		
● To build awareness of ethical considerations in deep learning, including bias, fairness, and privacy concerns.		
● To foster responsible AI practices through the study of explainable AI, federated learning, and ethical guidelines for AI research.		

Syllabus:

Module - 1 Introduction to Deep Learning	**No. of hours**
	06
Overview of machine learning and deep learning, History and Evolution of Deep Learning. Introduction to Neural networks: Perceptron, Multilayer Perceptron, Backpropagation, Shallow neural networks, deep neural networks, Optimizers: Gradient Descent (GD), Momentum-Based GD, Nesterov Accelerated GD, Stochastic GD, AdaDelta, AdaGrad, RMSProp, Adam, Loss Functions, Introduction to CUDA.	

Module - 2 Convolutional Neural Networks	**No. of hours**
	06
Introduction to Convolutional Neural Networks, Layers in CNN, Types of convolutions, Regularization Techniques: L1/L2 regularization, dropout, Early stopping, Data augmentation, A typical CNN structure, Standard CNN models: AlexNet, VGGNet, GoogLeNet, ResNet, Inception.	

Module - 3 Unsupervised Learning and Generative Modeling with Autoencoders and GANs	**No. of hours**
	06

Autoencoder (Types: Linear, CNN-based), Training Autoencoders, Variational Autoencoders (VAE), Introduction to GAN, GAN Architecture (Generator, Discriminator), Applications.

Module - 4 Sequence-to-sequence models	**No. of hours**
	06
Introduction to Recurrent Neural Networks and their structure, Challenges in RNN (Vanishing and Exploding Gradients), Long-short term memory (LSTM), Gated Recurrent Unit (GRU), Attention mechanism in RNNs, and Transformers.	

Module - 5 Deep Learning SDKs and Ethical Considerations	**No. of hours**
	06
Overview of deep learning SDKs, Importance of optimization and deployment in deep learning, Intel's OpenVINO Toolkit, Qualcomm's SNPE, NVIDIA TensorRT.	

Lab Programs	**[30 Hours]**
1	Write a program to implement a classification problem using an Artificial Neural Network (ANN). (a) Build the model using varying numbers of hidden layers and neurons in each hidden layer. (b) Use different optimizers and compare their accuracies. (c) Apply dropout and compare the results with those obtained without using dropout by varying the dropout rate. (d) Plot graphs to visualize all the results obtained from the above steps (e) Analyze the impact of hyperparameter tuning (e.g., learning rate, batch size) on model performance.
2	Create a program to implement a Convolutional Neural Network (CNN) for the given dataset. (a) Design a customized CNN model. (b) Experiment with and without Batch Normalization and compare their accuracy. (c) Evaluate the results against at least three different CNN variants. (d) Visualize the results by plotting graphs (e) Investigate techniques for handling potential dataset imbalances or applying advanced data augmentation
3	Implement a program to build Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) models for the given dataset. (a) Create customized RNN and LSTM architectures. (b) Experiment with variants of RNN and LSTM and compare their performance in terms of accuracy. (c) Analyze and compare the results obtained from RNN and LSTM models. (d) Plot graphs to visualize the loss and accuracy metrics.

4	Develop a program to implement a Convolutional Neural Network (CNN) for the given dataset. (a) Construct the CNN model and train it on the dataset. (b) Evaluate the model performance and interpret the results using explainable AI techniques: LIME, SHAP, SmoothGRAD, Occlusion, Saliency Maps. (c) Visualize the interpretations through graphical representations.
5	Speed up deep learning inference using NVIDIA TensorRT - Classification

Course outcomes
CO1: Apply neural networks to perform classification tasks
CO2: Implement and Apply Convolutional Neural Networks (CNNs) for image classification.
CO3: Apply VAE and GAN for generating the data, particularly for image data.
CO4: Utilize Recurrent Neural Network (RNN) architectures for sequence modeling and text generation.
CO5: Deploy and optimize deep learning models using SDKs such as OpenVINO, SNPE, and TensorRT on diverse hardware platforms, while evaluating and addressing ethical challenges in Deep Learning

Text books	
1.	Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. Deep learning. MIT Press, 2016, ISBN : 9780262035613.
2.	Zhang, Aston, et al. "Dive into deep learning." Cambridge University Press, 2023, ISBN-13 : 9781009389433.

Reference books	
1.	CS231n: Deep Learning for Computer Vision, Stanford University.
2.	CS6910: Deep Learning, IIT Madras.
3.	Practical Deep Learning, Fast.ai (https://course.fast.ai/)

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Lesson plan (Template –2)

Name of the Programme:	**B.Tech (Hons.)**	
Title of the Course:	**Deep Learning**	
Course code	**CS2228**	
Semester:	**IV**	
Names of the Course Instructor(s)		
Course credit	**No. of hours per week (2+0+2)**	**Total no. of Teaching hours**
03	**04 Hours**	**60 Hours**

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/reference
1	1	Introductory class, Unit 1: Overview of machine learning and deep learning and History and Evolution of Deep Learning	Interactive lecture + Whiteboard	Textbook1: Chapter 1 Textbook2: Chapter 1
2		Lab 1	Colab Notebook + Project	
3		Lab 1	Colab Notebook + Project	
4		Introduction to Neural networks: Perceptron, Multilayer Perceptron	Whiteboard for problem solving (worksheets), Videos for Perceptron, Colab for Multilayer perceptron	Textbook1: Chapter 6 Textbook2: Chapter 3
5		Backpropagation	Whiteboard for derivation and problem solving (worksheets)	Textbook1: Chapter 1 CS231n: Chapter 3
6		Lab 1	Colab Notebook + Project	
7		Lab 1	Colab Notebook + Project	
8		Shallow neural networks, deep	Interactive lecture + Whiteboard	Textbook1: Chapter 6

		neural networks		Textbook2: Chapter 5
9		Optimizers: Gradient Descent (GD), Momentum-Based GD, Nesterov Accelerated GD, Stochastic GD,	Whiteboard for derivation and problem solving (worksheets)	Textbook1: Chapter 8 Textbook2: Chapter 11
10		Lab 1	Colab Notebook + Project	
11		Lab 1	Colab Notebook + Project	
12		Optimizers: AdaDelta, AdaGrad, RMSProp, Adam	Whiteboard for derivation and problem solving (worksheets)	Textbook1: Chapter 8 Textbook2: Chapter 11
13		Lab 1	Colab Notebook + Project	
14		Lab 1	Colab Notebook + Project	
15		Unit 2: Introduction to Convolutional Neural Networks	Interactive lecture + Whiteboard	Textbook1: Chapter 9 Textbook2: Chapter 6
16		About the Images, channels	Interactive lecture + Whiteboard	Textbook1: Chapter 9 Textbook2: Chapter 6
17		Lab 2	Colab Notebook + Project	
18		Lab 2	Colab Notebook + Project	
19		Layers in CNN	Problem solving using worksheets	Textbook1: Chapter 9 Textbook2: Chapter 6
20	2	Types of convolutions, Regularization Techniques: L1/L2 regularization	Interactive lecture + Whiteboard + Problem solving using worksheets	Textbook1: Chapter 9 Textbook2: Chapter 6
21		Lab 2	Colab Notebook + Project	
22		Lab 2	Colab Notebook + Project	
23		dropout, Early stopping, Data augmentation	Interactive lecture + Whiteboard	Textbook1: Chapter 7 Textbook2: Chapter 4
24		Lab 2	Colab Notebook + Project	

25		Lab 2	Colab Notebook + Project	
26		A typical CNN structure, Standard CNN models: AlexNet, VGGNet	Colab Notebook	Textbook1: Chapter 9 Textbook2: Chapter 6
27		GoogLeNet, ResNet, Inception	Colab Notebook	Textbook1: Chapter 12 Textbook2: Chapter 7
28		Lab 3	Colab Notebook + Project	
29		Lab 3	Colab Notebook + Project	
30		Unit 3: Autoencoder	Interactive lecture + Whiteboard	Textbook1: Chapter 14 Textbook2: Chapter 16
31		Types: Linear, CNN-based	Interactive lecture + Whiteboard	Textbook1: Chapter 14 Textbook2: Chapter 16
32		Lab 3	Colab Notebook + Project	
33		Lab 3	Colab Notebook + Project	
34		Training Autoencoders	Interactive lecture + Whiteboard	Textbook1: Chapter 4-5 Textbook2: Chapter 16
35	**3**	Variational Autoencoders (VAE)	Interactive lecture + Whiteboard	Textbook1: Chapter 20
36		Lab 3	Colab Notebook + Project	
37		Lab 3	Colab Notebook + Project	
38		Introduction to GAN, GAN Architecture (Generator, Discriminator),	Interactive lecture + Whiteboard	Textbook1: Chapter 20 Textbook2: Chapter 17
39		GAN and VAE Loss functions	Interactive lecture + Whiteboard	Textbook1: Chapter 20 Textbook2: Chapter 17
40		Hands-on on VAE and GAN	Colab Notebook + Project	
41		Lab 4	Colab Notebook + Project	XAI: LIME
42		Lab 4	Colab Notebook + Project	XAI: LIME

43	**4**	Unit 4: Sequence-to-Sequence Models	Interactive lecture + Whiteboard + Problem solving using worksheets	Textbook1: Chapter 10 Textbook2: Chapter 8
44		Introduction to Recurrent Neural Networks	Interactive lecture + Whiteboard + Problem solving using worksheets	Textbook1: Chapter 10 Textbook2: Chapter 8
45		Lab 4	Colab Notebook + Project	XAI: SHAP, SmoothGRAD
46		Lab 4	Colab Notebook + Project	XAI: SHAP, SmoothGRAD
47		RNN Structure and Challenges in RNN	Interactive lecture + Whiteboard + Problem solving using worksheets	Textbook1: Chapter 10 Textbook2: Chapter 8
48		Long-short term memory (LSTM),	Interactive lecture + Whiteboard + Problem solving using worksheets	Textbook1: Chapter 6 - 8 Textbook2: Chapter 9
49		Lab 4	Colab Notebook + Project	XAI: Occlusion, Saliency Maps
50		Lab 4	Colab Notebook + Project	XAI: Occlusion, Saliency Maps
51		Gated Recurrent Unit (GRU),	Interactive lecture + Whiteboard + Problem solving using worksheets	Textbook1: Chapter 10 Textbook2: Chapter 9
52		Attention mechanism in RNNs	Interactive lecture + Whiteboard + Problem solving using worksheets	Textbook1: Chapter 12 Textbook2: Chapter 10
53		Transformers.	Interactive lecture + Whiteboard + Problem solving using worksheets	Textbook2: Chapter 10
54		Lab 5	Colab Notebook + Project	
55		Lab 5	Colab Notebook + Project	
56		Intel's OpenVINO	Colab Notebook	Dell web content
57		TensorRT	Colab Notebook	NVIDIA web content
58		Lab 5	Colab Notebook + Project	

59	**5**	Lab 5	Colab Notebook + Project	
60		Capstone Project Development.		

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1 - Lab 1, Review 1 & Quiz	2+8+10 marks out of 70
CIE-2 - Theory	25 marks out of 70
CIE-3 - Research paper in IEEE format + Plagiarism report - Working code - Final Project Submission through presentation - Lab 2, Lab 3, Lab 4, and Lab 5	10 marks out of 70 3 marks out of 70 4 marks out of 70 2+2+2+2 marks out of 70

Semester End Exams (SEE): 30%	
Mode of Exam	Theory
Weightage of exam	30%

Course Outcomes- Programme Outcomes Matrix

Program Outcome Course Outcome		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12
CO 1	CO1: Apply neural networks to perform classification tasks	3	2	3		3	1						

CO 2	CO2: Implement & Apply Convolutional Neural Networks (CNNs) for image classification.	3	2	3		3							
CO 3	CO3: Apply VAE and GAN for generating the data, particularly for image data.	3	2	3	2	2	1						
CO 4	CO4: Utilize Recurrent Neural Network (RNN) architectures for sequence modeling and text generation.	3	2	3	2	2	1	1	1				
CO 5	CO5: Deploy and optimize deep learning models using SDKs such as OpenVINO, SNPE, and TensorRT on diverse hardware platforms, while evaluating and addressing ethical challenges in Deep Learning	3	2	3	2	2	1	1	1				

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		**20**
CIE-1 Components		**Rubrics for Assessment**
CIE-1	Lab 1	2 marks
	Quiz	10 marks ***(Refer Table 1)***
	Review 1	08 marks ***(Refer Table 2)***

Total Marks		**25**
CIE-2 Components		**Rubrics for Assessment**
CIE-2	Theory	The test will be conducted for 25 marks and will be evaluated according to the Scheme of Evaluation prepared by the team of course faculty.

Total Marks		**25**
CIE-3 Components		**Rubrics for Assessment**
CIE-3	Research paper in IEEE format + Plagiarism report	10 marks out of 70 ***(Refer Table 3)***
	Working code	3 marks out of 70 ***(Refer Table 4)***
	Final Project Submission through presentation	4 marks out of 70 ***(Refer Table 5)***
	Lab2, Lab 3, Lab 4, and Lab 5	2+2+2+2 = 8 marks out of 70 ***(Refer Table 6)***

Table 1: Quiz Rubric Statement (10 Marks)

This rubric is for evaluating individual quiz performance in the Deep Learning course.

Quiz	The quiz will be conducted for a total of 10 marks. Each question carries variable weightage based on difficulty level / cognitive level. There is no negative marking for wrong answers.

Table 2: Deep Learning Project Review 1 Rubric (8 Marks)

This rubric evaluates the student's performance in Review 1, which includes Problem Definition and Literature Review.

Component	Criteria	Excellent (8)	Good (5)	Needs Improvement (2)	Marks
Research Problem Identification	Clearly defined, well-scoped DL problem relevant to real-world or research domain.	The problem is innovative, well-articulated, and feasible.	The problem is understandable but lacks depth or clarity.	Vague, broad, or disconnected from DL context.	/2
Objectives	Objectives align with the problem and are SMART (Specific, Measurable, Achievable, Relevant, Time-bound).	Clear, well-structured objectives aligned with the problem.	Objectives exist but not fully SMART or well-aligned.	Objectives missing, vague, or not clearly related.	/1
Literature Review Coverage	At least 6–8 recent and relevant papers (preferably IEEE/ACM) are cited.	Comprehensive and current review covering gaps and trends.	Moderate coverage; few recent or relevant works included.	Limited or outdated literature; lacks research relevance.	/2
Critical Analysis	Comparison of prior	Deep insights, gap	Some analysis, but lacks depth	Mostly descriptive; no critical	/2

	work, identification of research gaps, justification of chosen approach.	identification, and synthesis evident.	or connection to the problem.	evaluation or insights.	
Presentation & Formatting	The report is well-organized, properly formatted, and free from grammatical issues. Uses IEEE style.	Professional structure, error-free, and well formatted.	Acceptable formatting with few errors.	Poor formatting, errors, or missing IEEE structure.	/1

Table 3: Research Paper Submission Rubric (10 Marks)

Criteria	Excellent (9–10 M)	Very Good (7–8 M)	Good (5–6 M)	Needs Improvement (0–4 M)
Content Quality, IEEE Formatting & References, Plagiarism Report	Innovative idea, clear objectives, DL theory aligned, IEEE format, <10% plagiarism.	Mostly complete, minor format issues, 10–15% plagiarism.	Basic idea, limited insight, some citation issues, 15–25% plagiarism.	Lacks structure/originality, >25% plagiarism.

Table 4: Code Evaluation Rubric (3 Marks)

This rubric is used for evaluating short coding assignments or lab submissions.

Component	Excellent (3 M)	Good (2 M)	Needs Improvement (1–2 M)
Functionality	Code runs correctly, handles edge cases, and produces expected results	Code runs with minor issues or limited testing	Code has significant errors or does not run

Code Quality	Follows good practices, well-structured, with meaningful variable names	Acceptable structure, some best practices missing	Poorly structured, unclear naming, or hard to read
Use of DL Libraries	Appropriate use of relevant libraries (e.g., TensorFlow, PyTorch, Keras)	Limited use of relevant libraries	Irrelevant or missing DL components
Comments & Documentation	Well-commented, README or inline explanation provided	Partial documentation or comments	No comments or documentation
Reproducibility	Code is easy to run with clear setup instructions or environment file	Code runs with some effort	Code hard to reproduce or lacks setup details

Table 5: Final Project Submission through Presentation (4 Marks)

This rubric evaluates the final project presentation based on clarity, technical depth, results, and communication.

Criteria	Excellent (4 M)	Good (3 M)	Needs Improvement (1–2 M)
Problem Understanding	Clearly explains the problem, motivation, and relevance	Basic explanation with some gaps	Unclear or lacks relevance
Model & Methodology	Accurate description of DL model, architecture, and tools used	Partial explanation with minor technical gaps	Superficial or incorrect methodology
Results & Insights	Strong results with visualizations; well-interpreted	Acceptable results; some insights	Weak or missing results
Presentation Skills	Confident, clear, and time-managed delivery	Some hesitation or time management issues	Unclear, rushed, or poorly managed

Q&A Handling	Accurate and thoughtful responses	Partially correct responses	Unable to address questions

Final Project Submission Guidelines

Course: Deep Learning
Components: IEEE-format Report + GitHub Code Repository

1. Project Report Guidelines (IEEE Format)

• Format: The report must strictly follow IEEE Conference paper format, available at: https://www.ieee.org/conferences/publishing/templates.html
 - Use A4-size, two-column layout, and Times New Roman font (size 10).
 - Maintain IEEE structure: Abstract, Introduction, Related Work, Methodology, Results, Conclusion, References.

• Length: 6 to 8 pages (excluding references).

• Plagiarism Check: Must be less than 15% (use Turnitin, Grammarly, or institutional tool).

• AI Detection Score: AI-generated content must be below 15% (e.g., GPTZero, Copyleaks).

• Citations: IEEE citation style with numbering [1], [2], etc.
 - Minimum of 6–8 scholarly references (preferably IEEE, ACM, Springer, Scopus).

2. Code Submission Guidelines (GitHub Repository)

• Platform: Upload code to a GitHub repository.

• Recommended Structure:
```
├── README.md (project title, abstract, setup instructions)
├── requirements.txt or environment.yml (dependencies)
├── /data (sample data or dummy files only)
├── /notebooks or /src (scripts or Jupyter Notebooks)
├── /results (output images, charts, logs, etc.)
└── report.pdf (final IEEE report)
```

• README File:
 - Include project title, abstract, setup instructions, and usage guide.

- Link to the final PDF report.

• Reproducibility:
- Code must run from scratch with minimal effort.
- Include instructions and any pre-trained models or data links.
- Output must align with reported results.

3. Submission Checklist

Item	Requirement
Report Format	IEEE 2-column conference format
Page Limit	6–8 pages
Plagiarism	Less than 15%
AI Content Percentage	Less than 15%
References	At least 15–20 scholarly papers
GitHub Repository	Public and accessible
README with Instructions	Complete and clearly written
Reproducible Results	Code runs without errors

4. Deadline & Submission Instructions

• Deadline: 5 days before the last working day.

• Submission Mode: LMS / Google Form / Email
• Files to Submit:
1. Final IEEE-format Report in PDF
2. GitHub Repository Link

Table 6: Lab Execution Rubric (10 Marks)

Sl. No	Criteria	Measuring Methods	Excellent (10 M)	Good (8 M)	Poor (2 M)
1	Understanding of DL Problem Statements	Observations	In-depth understanding of DL concepts; suitable model chosen.	Intermediate level of understanding	Basic understanding; model acceptable or poor.

2	Code Execution & Efficiency	Observations	Optimized code with appropriate libraries and correct pipeline.	Correct but unoptimized or incomplete/incorrect code.	
3	Results and Documentation	Observations	Includes visualizations, well-commented code, and outputs.	Partial visual output and documentation.	Missing visual output and documentation.

Signature of the Course Faculty